

OptiCrop

Smart Agricultural Production Optimization Engine

Project Documentation

1. Introduction

Project Title

OptiCrop – Smart Agricultural Production Optimization Engine

Repository : <https://github.com/Madhan-9/OptiCrop>

Team Members

Team Member	Role
Tadisetti Nagasiri	Team Leader
NAGA PRANEETHA KANNA	Team member
Munishwar Ande	Team member
Harshithreddy Chintalapudi	Team member
Madhan Sankarasetty	Team member

Mentor: No Mentor Assigned

2. Project Overview

Purpose

OptiCrop is a machine learning-based agricultural recommendation system designed to optimize crop production using environmental and soil parameters. It integrates ML algorithms with a Flask web application to provide intelligent crop recommendations based on key environmental factors, helping farmers, agricultural researchers, agribusiness companies, and policymakers make data-driven decisions that improve productivity, sustainability, and resource utilization.

Features

- Smart crop recommendation based on soil nutrients (N, P, K) and environmental parameters
- Crop suitability and environmental assessment for a given set of conditions
- Support for agricultural research and policy planning through data-driven pattern analysis
- Comparison of multiple ML algorithms (KNN, Logistic Regression, Decision Tree, Random Forest, K-Means) to select the best-performing model

- Interactive, responsive web interface for entering parameters and viewing instant predictions
- Trained model persistence using Pickle (.pkl) for fast reuse during prediction

3. Architecture

Frontend

Built using HTML, CSS, and Bootstrap for a responsive, user-friendly interface. Input forms collect agricultural parameters — Nitrogen (N), Phosphorous (P), Potassium (K), Temperature, Humidity, pH, and Rainfall — and prediction buttons trigger instant crop recommendations, with results displayed dynamically on the results page.

Backend

Built using Flask (Python) with modular functions handling data preprocessing and cleaning, model training and evaluation, crop recommendation prediction, and input validation. The backend receives user input, passes it to the trained ML model, and returns the predicted crop recommendation.

Machine Learning / Data Layer

The dataset containing soil nutrients and environmental parameters is preprocessed (handling missing values, feature scaling, categorical transformations) and used to train and compare multiple models (K-Nearest Neighbors, Logistic Regression, Decision Tree, Random Forest, K-Means). The best-performing model is serialized using Pickle (.pkl) and loaded by the backend at prediction time.

Note: This project uses a trained ML model file rather than a traditional database (e.g., MongoDB). If a database is used for storing user inputs, history, or logs, add its schema and details here.

4. Setup Instructions

Prerequisites

- Python 3.10+
- pip / Anaconda Navigator
- Jupyter Notebook or Spyder
- Visual Studio Code (or equivalent IDE)
- Git

Required Libraries

- Flask
- Scikit-learn
- Pandas & NumPy
- Matplotlib & Seaborn
- SciPy
- Pickle

Installation

1. Clone the repository:

```
git clone https://github.com/Madhan-9/OptiCrop.git cd OptiCrop
```

2. Create and activate a virtual environment:

```
python -m venv venv venv\Scripts\activate # Windows source venv/bin/activate #
macOS/Linux
```

3. Install dependencies:

```
pip install -r requirements.txt
```

Note: Confirm the exact requirements.txt / environment variable names in the repo and update this section accordingly.

5. Folder Structure

The structure below is a suggested Flask project layout. Replace with the actual structure from the repository.

Project Root

```
OptiCrop/ ├── static/           # CSS, JS, images ├── templates/       # HTML templates
(Jinja2) ├── model/             # Trained model (.pkl) files ├── dataset/
# Agricultural dataset(s) ├── app.py           # Flask application entry point ├──
requirements.txt └── README.md
```

6. Running the Application

Start the Flask application locally:

```
python app.py
```

By default, the app will be available at:

```
http://127.0.0.1:5000/
```

7. API Documentation

The endpoints below are inferred from the project description. Verify against app.py and update method names, parameters, and example responses as needed.

POST /predict

Accepts agricultural parameters and returns the recommended crop.

Request parameters:

- nitrogen (N)
- phosphorous (P)
- potassium (K)
- temperature
- humidity
- ph
- rainfall

Example response:

```
{  "recommended_crop": "rice",  "confidence": 0.92 }
```

8. Authentication

No authentication/authorization mechanism is described in the current project scope. If login, sessions, or tokens are implemented, document them here (e.g., Flask-Login, JWT, session cookies).

9. User Interface

Add screenshots or GIFs of the input form, prediction results page, and any dashboards here.

10. Testing

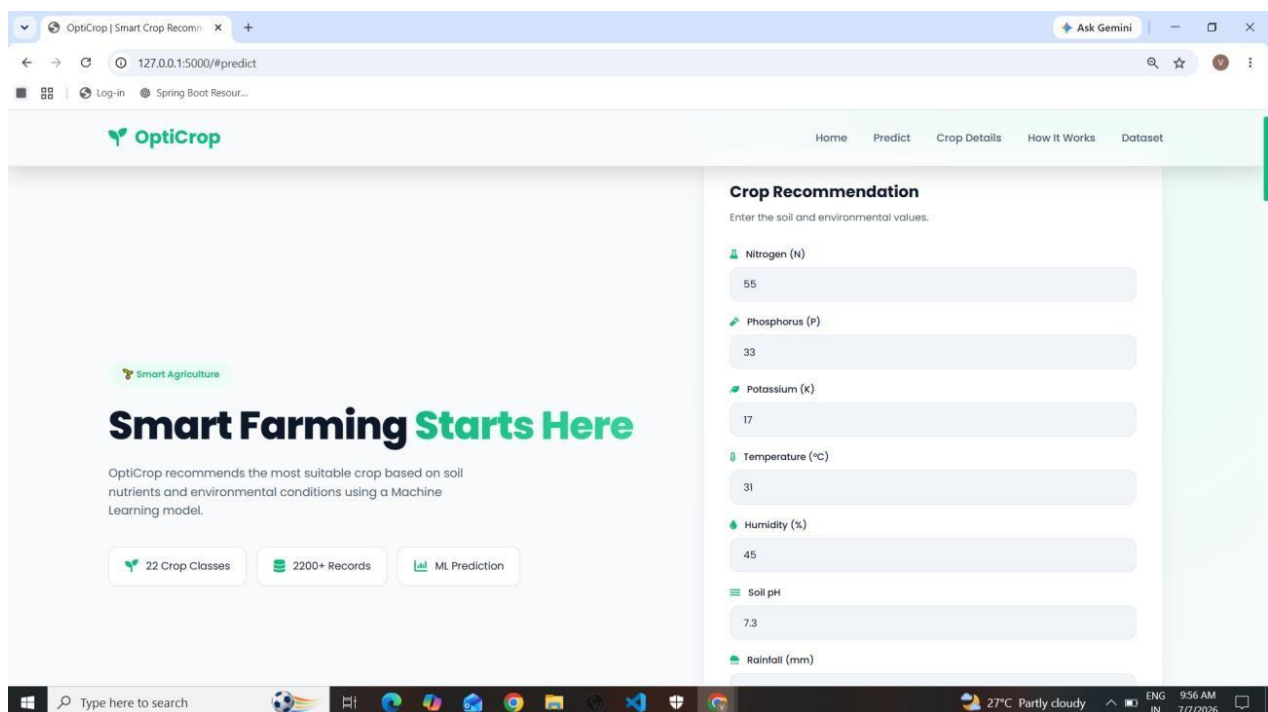
Testing is performed to ensure accurate crop recommendations and a stable web application:

- Model accuracy evaluation across multiple algorithms (KNN, Logistic Regression, Decision Tree, Random Forest, K-Means)
- Flask development server testing
- Browser-based manual testing of input forms and prediction results

11. Screenshots or Demo Link

: <http://127.0.0.1:5000/>

Screenshot: example-1:



OptiCrop | Smart Crop Recommendation

Ask Gemini

127.0.0.1:5000/#predict

Log-inSpring Boot Resour...

OptiCrop

HomePredictCrop DetailsHow It WorksDataset

PREDICTION RESULT

Recommended Crop

Based on the soil nutrients and environmental conditions, OptiCrop recommends the following crop.



Best Recommendation

pigeonpeas

Pigeon pea is a drought-resistant pulse crop widely cultivated in tropical and subtropical regions.

PREDICTION CONFIDENCE

25%

↑

OptiCrop | Smart Crop Recommendation

Ask Gemini

127.0.0.1:5000/#details

Log-inSpring Boot Resour...

OptiCrop

HomePredictCrop DetailsHow It WorksDataset

CROP INFORMATION

Complete Crop Details

Detailed agricultural information for the recommended crop.

Season

Kharif

Harvest Time

Soil Type

Red and Black Soil

Water Requirement

Moderate

Fertilizer

Phosphorus-based Fertilizer

Market Demand

High

Suitable Regions

Maharashtra,Karnataka,Gujarat

Common Uses

↑

OptiCrop | Smart Crop Recommendation

Ask Gemini

127.0.0.1:5000/#workflow

Log-inSpring Boot Resour...

OptiCrop

HomePredictCrop DetailsHow It WorksDataset

WORKFLOW

How OptiCrop Works

A simple four-step process to recommend the most suitable crop.

1

Enter Soil Data

Provide Nitrogen, Phosphorus, Potassium, pH, Temperature, Humidity and Rainfall.

2

Data Processing

Flask prepares the data and sends it to the trained Machine Learning model.

3

Prediction

The ML model predicts the most suitable crop for your conditions.

4

Recommendation

View complete crop information, growing conditions and cultivation tips.

127.0.0.1:5000/#workflow

127.0.0.1:5000/#dataset

Model Information

27°C Partly cloudyENG IN9:57 AM7/7/2026

OptiCrop | Smart Crop Recommendation

Ask Gemini

127.0.0.1:5000/#dataset

Log-inSpring Boot Resour...

OptiCrop

HomePredictCrop DetailsHow It WorksDataset

DATASET

Model Information

The recommendation model has been trained using agricultural soil and climate data.

2200+

Training Records

22

Crop Classes

7

Input Features

Random Forest

Machine Learning Model

OptiCrop

127.0.0.1:5000/#datasetSmart Crop Recommendation System

Quick Links

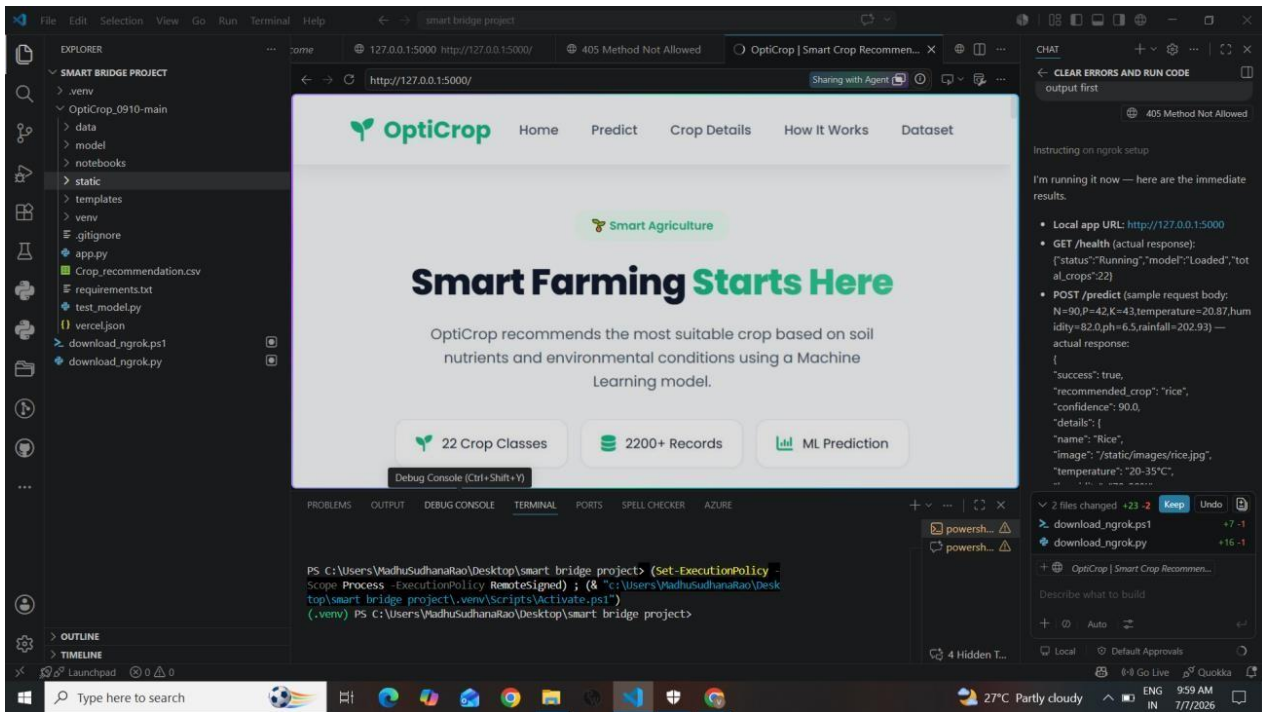
Home

Prediction

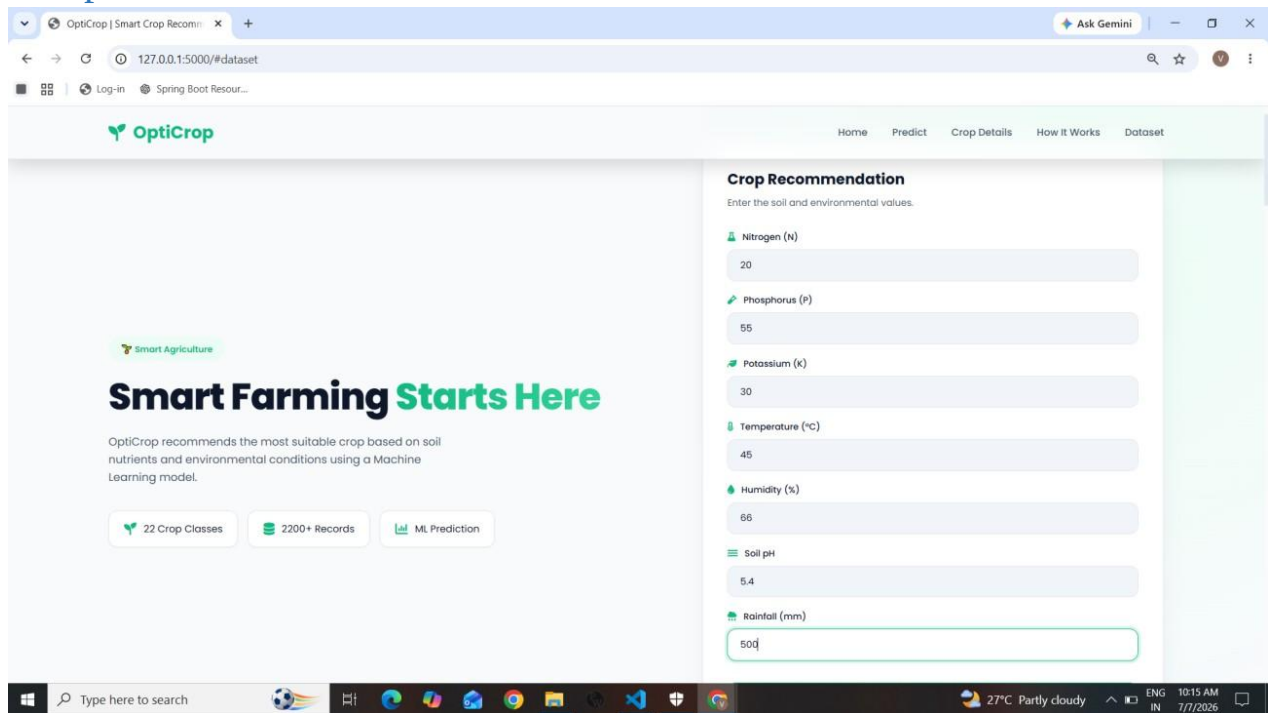
Technologies

Python

Flask



example-2:



OptiCrop | Smart Crop Recommendation

Ask Gemini

127.0.0.1:5000/#dataset

Log-inSpring Boot Resour...

OptiCrop

HomePredictCrop DetailsHow It WorksDataset

PREDICTION RESULT

Recommended Crop

Based on the soil nutrients and environmental conditions, OptiCrop recommends the following crop.



Best Recommendation

pigeonpeas

Pigeon pea is a drought-resistant pulse crop widely cultivated in tropical and subtropical regions.

PREDICTION CONFIDENCE

44 %

↑

OptiCrop | Smart Crop Recommendation

Ask Gemini

127.0.0.1:5000/#details

Log-inSpring Boot Resour...

OptiCrop

HomePredictCrop DetailsHow It WorksDataset

CROP INFORMATION

Complete Crop Details

Detailed agricultural information for the recommended crop.

**Season**
Kharif

**Harvest Time**

**Soil Type**
Red and Black Soil

**Water Requirement**
Moderate

**Fertilizer**
Phosphorus-based Fertilizer

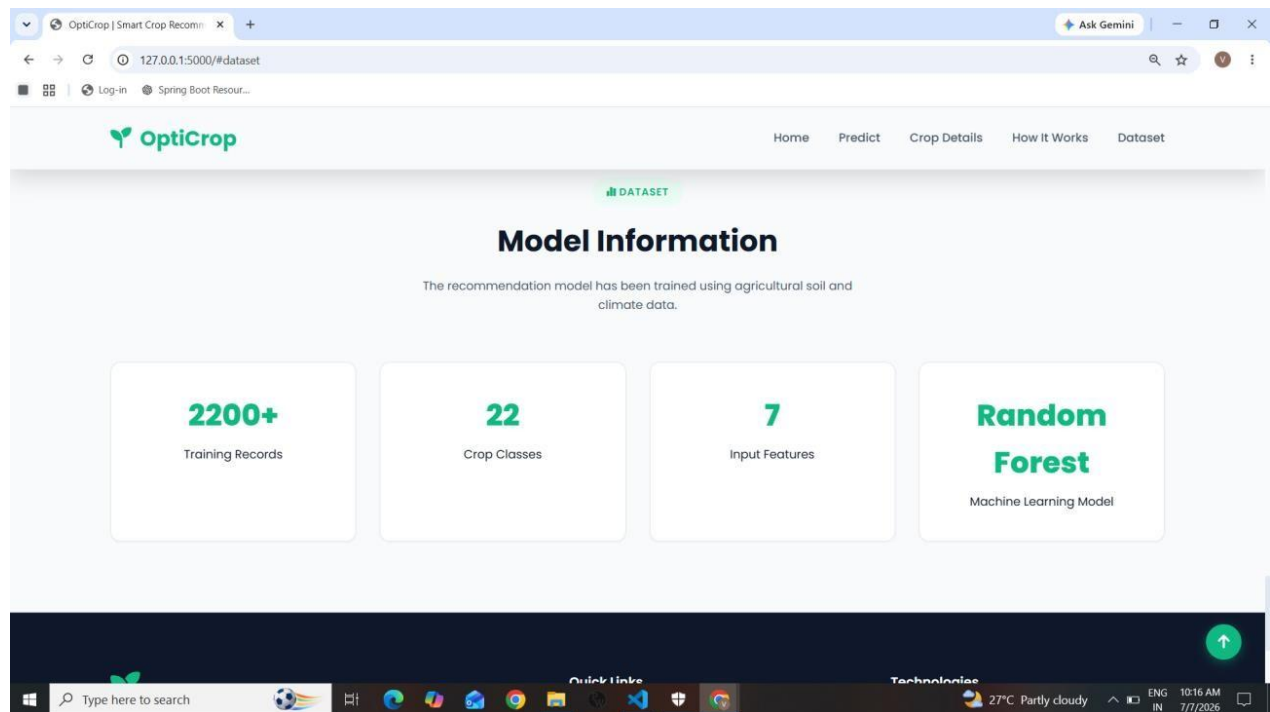
**Market Demand**
High

**Suitable Regions**
Maharashtra,Karnataka,Gujarat

**Common Uses**

↑

127.0.0.1:5000/#details



12. Known Issues

Document any known bugs or limitations here, e.g., dataset coverage limited to specific crops/regions, no database persistence, no user authentication yet.

13. Future Enhancements

- Add a database layer to store user queries and recommendation history
- Introduce user authentication and personalized dashboards
- Expand the dataset to cover more crops and regional variations
- Deploy the application to a cloud platform for public access
- Add visual analytics/dashboards for researchers and policymakers